DATABRICKS · HEALTHCARE / ICU OPERATIONS

# ICU Step-Down Readiness

Turning scarce ICU capacity into a decision advantage — an AI-assisted way to see, every shift, which patients are most ready to safely leave intensive care.

Lakeflow | Unity Catalog | Lakebase | Mosaic AI + Foundation Models | Genie | Databricks Apps

THE PROBLEM

## A blocked ICU bed is a clinical and financial bottleneck

### 70–90%

Typical ICU occupancy — one blocked bed delays a critical admission

### Manual

Step-down decisions are subjective and vary by provider and shift

### "To be safe"

Clinicians hold patients longer than needed, lengthening stays

The question every intensivist asks on rounds — *“who can safely move to step-down today?”* — has no consistent, data-driven answer.

WHY IT MATTERS BEYOND THE UNIT

# ICU length of stay is a margin, benchmarking, and market-share lever

### Payer & margin risk

Under DRG-fixed and value-based contracts, every avoidable ICU day is **unreimbursed cost**, and ICU bounce-backs feed readmission penalties and quality scores (CMS HRRP, star ratings) that payers price on.

### Peer benchmarking

ICU LOS, occupancy, and throughput are ranked against peer hospitals (Vizient, CMS, Leapfrog). Being an **LOS outlier** is a board- and payer-visible signal of inefficiency.

### Competitive position & growth

A full ICU means **diverted ambulances, declined tertiary referrals, and postponed elective surgery** — lost revenue, lost referral relationships, and lost market share. Freed capacity is growth capacity.

The intensivist's daily step-down call, aggregated across the unit, is what a **CFO, CMO, and market strategist** see as reimbursement exposure, a benchmark percentile, and the ability to say yes to the next transfer.

THE OUTCOME WE SELL

## Free ICU capacity without compromising safety

**10–15%**
Target reduction in ICU length of stay

**$2–5k**
Saved per accelerated discharge-day

**+beds**
More ICU capacity for emergent admissions

**Consistency**
Evidence-informed decisions across shifts

For a 30-bed ICU, a 0.8-day average LOS reduction is on the order of **hundreds of additional patient-days of capacity per year**.

WHAT WE BUILT

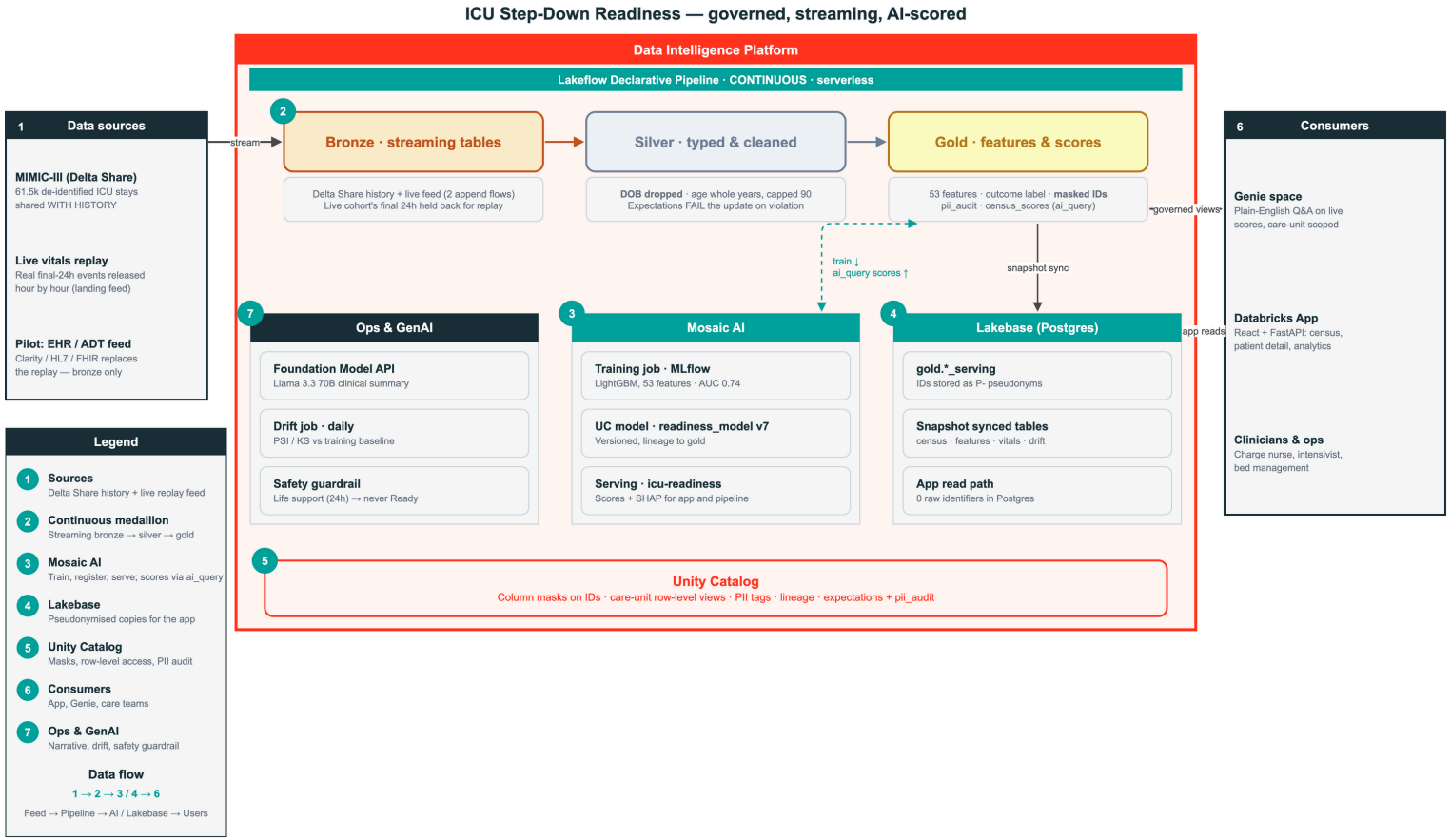
# One governed journey — raw data to a clinician's screen

**MIMIC-III (Delta Share)** → **Lakeflow medallion** → **Unity Catalog** →
**Lakebase serving** → **ML + Gen AI** → **Genie** → **Databricks App**

Integrated, not siloed: the same governed gold tables feed the operational app, the model, and natural-language querying — one lineage, one source of truth.

ARCHITECTURE

# Governed, continuously streaming, AI-scored — on one platform




Source data streams in continuously; silver drops date of birth; gold carries masked identifiers and live model scores; the app reads pseudonymised Lakebase copies; Unity Catalog governs every layer. Editable source: `deck/architecture.drawio`.

THE PRODUCT

## A ranked ICU census the care team acts on

- **Census dashboard** — every current patient ranked by a relative readiness index (re-scored within minutes as new vitals stream in), color-banded Ready / Borderline / Not-ready.
- **Patient detail** — the factors driving each recommendation, vital-sign trends, and a plain-language **AI clinical summary**.
- **Analytics** — capacity and outcome trends, with honest model performance (v7: AUC 0.74, 0.64 readmission-only).

Live: `icu-step-down-7474645901826067.aws.databricksapps.com`

Example — ranked census

🟢 **92** · Ready · off pressors 36h, GCS 15

🟢 **78** · Ready · extubated, stable VS

🟡 **54** · Borderline · lactate ↑

🔴 **18** · Not ready · on vent + pressors

FOR THE CMIO & ICU MEDICAL DIRECTOR

# Built to be trusted, governed, and operated

### Governed by design

Unity Catalog across the medallion — DOB dropped, identifiers pseudonymized via UC column masks, PII classification tags, least-privilege grants, full bronze→gold lineage with continuous audit.

### Live-fed & operationally served

Continuous Lakeflow streaming pipeline — new vitals in bronze in seconds, scores in ~2 min; pseudonymised gold synced to Lakebase (Postgres) for low-latency app reads; model served on Mosaic AI; narrative via Foundation Model API.

### Reproducible

Everything deploys via Databricks Asset Bundles; 199 backend + 38 frontend + 71 pipeline/ML unit + 28 live integration tests, run output committed.

### Safe by framing

Public de-identified MIMIC; a relative decision-support index, not an automated discharge gate.

MODEL HONESTY

## Decision support, not a discharge switch

### AUC 0.74

95% CI 0.71–0.76 on patients who left the ICU alive — the only patients a step-down decision is made for. Flags not-safe step-downs at ~5× the base rate

### Readmission-only 0.64

72h bounce-back alone is genuinely hard from structured data. We say so, rather than hide it

### Index, not %

Relative readiness rank within the current census — never a "probability of safe discharge"

Explainable: SHAP factors in clinical language; known risks always shown as risks, and a patient on vasopressors or a ventilator in the last 24h is never banded Ready. We retired an earlier 0.89: it counted patients who died in the ICU, whose final day is easy to recognise. The next lift comes from your data, notes and validation with your clinicians.

PROJECT PLAN

# Trained on your history, proven on your unit — before anyone relies on it

Weeks 1–4 · Train on your … | Weeks 5–8 · Silent pilot | Weeks 9–16 · Clinician pilot | Week 17+ · Scale

PHASE 1 · WEEKS 1–4

## Train on your historical data

- Point the same governed pipeline at your EHR feed — 2–3 years of your ICU stays
- Retrain on your patients, practice and outcomes
- ICU Medical Director co-defines features, guardrails and the “safe step-down” label

**Exit gate:** discrimination and calibration on your held-out stays, checked by unit and patient subgroup

PHASE 2 · WEEKS 5–8

## Silent pilot

- Live scores every shift, *not shown* to clinicians
- Compared with the step-down calls clinicians actually made, and 72h outcomes
- Validates the numbers on today's patients, not just history

**Exit gate:** live performance matches retrospective; no safety signal in Ready-banded patients

PHASE 3 · WEEKS 9–16

## Clinician pilot — one ICU

- Ranked census used on rounds; the decision stays with the clinician
- Clinicians record agree / disagree and why
- Weekly review of flagged vs actual bounce-backs builds trust and tunes thresholds

**Exit gate:** ICU LOS reduced with safe-transfer and 72h bounce-back rates held or improved

PHASE 4 · WEEK 17+

## Decide & scale

- Go / no-go with clinical governance
- Extend unit by unit on the same platform
- Drift monitoring and scheduled retraining

**Ongoing:** performance reported to the clinical governance board

Clinicians see nothing until the model has been validated twice: first on your history, then silently on live patients. Trust is earned before the index is used on rounds.

THE ASK

## From prototype to a unit pilot

- **VP of Clinical Operations / CMO** — measured on ICU length of stay, bed occupancy, and emergent-admission wait times: sponsor the plan, with an 8-week clinician pilot in one ICU scored on LOS reduction with the safe-transfer rate held or improved.
- **ICU Medical Director** — measured on safe-transfer rate and 48–72h ICU bounce-back: co-define the readiness features and validate the index against your step-down criteria.
- **CMIO / VP Data & Analytics** — measured on time-to-value and governed reuse: point the same pipeline at your EHR feed — the architecture is unchanged.

Every ask maps to a number already on that leader's own scorecard. The capability is live today on Databricks; the next step is your data and your clinicians.